Independent Analytical Note

Derivation of the Moment of Inertia of a Solid Torus via Hollow Torus Layering

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Abstract

Calculating the moment of inertia (I) of a solid torus about its central axis of rotational symmetry is traditionally performed using triple integrals in toroidal or spherical-polar coordinate systems requiring Jacobian determinants. In this note, I propose a two-step layering method to derive the moment of inertia of a uniform solid torus about its primary axis of rotational symmetry. First, I compute the rotational inertia of a thin hollow torus (toroidal shell) of mass M using surface element integration. Subsequently, I treat the solid torus as a continuum of infinitely thin concentric hollow toroidal shells to perform a single-variable radial integration. This yields the standard result for a solid torus:

$$I_{\text{solid}} = M\left(R^2 + \frac{3}{4}r^2\right)$$

1 Part 1: Moment of Inertia of a Thin Hollow Torus

1.1 Geometric Setup & Surface Area Element

Let a thin hollow torus be formed by revolving a circle of radius r centered at a distance R from the axis of rotation ($R > r$). Parameterize the circular cross-section using the angle $\theta \in [0, 2\pi]$ measured relative to the horizontal plane.

- R: Major radius of the torus
- r: Minor radius (cross-sectional radius)
- M: Total mass of the hollow shell
- A: Total surface area of the hollow torus

The perpendicular distance from any point on the shell surface to the axis of rotation is:

$$x(\theta) = R + r\sin\theta$$

The differential arc length along the circular cross-section is $ds = r\,d\theta$. The circumferential perimeter of the ring at angle θ is $2\pi(R + r\sin\theta)$. Thus, the differential surface area dA is:

$$dA = 2\pi(R + r\sin\theta)\cdot r\,d\theta$$

Integrating dA over $\theta \in [0, 2\pi]$ yields the total surface area:

$$A = \int_0^{2\pi} 2\pi r(R + r\sin\theta)\,d\theta = 2\pi r(2\pi R) = 4\pi^2 Rr$$

1.2 Integration for the Hollow Torus

The differential mass dM of a strip at angle θ is:

$$dM = \frac{M}{A}dA = \frac{M}{4\pi^2 Rr} \cdot 2\pi r(R + r\sin\theta)\, d\theta = \frac{M}{2\pi R}(R + r\sin\theta)\, d\theta$$

The differential moment of inertia dI about the central axis is:

$$dI = dM \cdot [x(\theta)]^2 = dM(R + r\sin\theta)^2 = \frac{M}{2\pi R}(R + r\sin\theta)^3\, d\theta$$

Expanding the cubic term:

$$(R + r\sin\theta)^3 = R^3 + 3R^2 r\sin\theta + 3Rr^2\sin^2\theta + r^3\sin^3\theta$$

Integrating dI from $\theta = 0$ to 2π:

$$I_{\text{hollow}} = \frac{M}{2\pi R}\int_0^{2\pi} \left(R^3 + 3R^2 r\sin\theta + 3Rr^2\sin^2\theta + r^3\sin^3\theta\right) d\theta$$

Using symmetry properties of trigonometric integrals over a full period:

$$\int_0^{2\pi} \sin\theta\, d\theta = 0, \quad \int_0^{2\pi} \sin^3\theta\, d\theta = 0, \quad \int_0^{2\pi} \sin^2\theta\, d\theta = \pi$$

Evaluating the integral:

$$I_{\text{hollow}} = \frac{M}{2\pi R}\left[R^3(2\pi) + 3Rr^2(\pi)\right] = \frac{M}{2\pi R}\left(2\pi R^3 + 3\pi Rr^2\right)$$

$$I_{\text{hollow}} = MR^2 + \frac{3}{2}Mr^2 = M\left(R^2 + \frac{3}{2}r^2\right)$$

2 Part 2: Moment of Inertia of a Solid Torus via Shell Integration

2.1 Mass Distribution in Concentric Hollow Layers

Consider a solid torus of mass M, major radius R, and maximum minor radius r. We decompose the solid torus into concentric hollow toroidal shells of varying minor radius $x \in [0, r]$ and thickness dx. The differential volume element dV of a shell with minor radius x is:

$$dV = A(x)\, dx = (4\pi^2 Rx)\, dx$$

The total volume of the solid torus is $V = 2\pi^2 Rr^2$. The differential mass dM of a shell at radius x is:

$$dM = \frac{M}{V}dV = \frac{M}{2\pi^2 Rr^2} \cdot (4\pi^2 Rx\, dx) = \frac{2M}{r^2}x\, dx$$

2.2 Integration over the Solid Torus

Using the formula derived in Part 1 for a hollow torus of mass dM and minor radius x:

$$dI_{\text{solid}} = dM\left(R^2 + \frac{3}{2}x^2\right)$$

Substituting dM:

$$dI_{\text{solid}} = \left(\frac{2M}{r^2}x\, dx\right)\left(R^2 + \frac{3}{2}x^2\right) = \frac{2MR^2}{r^2}x\, dx + \frac{3M}{r^2}x^3\, dx$$

Integrating from $x = 0$ to $x = r$:

$$I_{\text{solid}} = \frac{2MR^2}{r^2}\int_0^r x\, dx + \frac{3M}{r^2}\int_0^r x^3\, dx$$

$$I_{\text{solid}} = \frac{2MR^2}{r^2}\left[\frac{r^2}{2}\right] + \frac{3M}{r^2}\left[\frac{r^4}{4}\right]$$

$$\mathbf{I_{\text{solid}} = MR^2 + \frac{3}{4}Mr^2 = M\left(R^2 + \frac{3}{4}r^2\right)}$$

3 Conclusion

In this note, I have presented a self-contained, two-step geometric layering approach to derive the moment of inertia of a uniform solid torus. By first evaluating the surface integral of a thin hollow toroidal shell, the problem of a 3D solid torus is effectively reduced to a straightforward single-variable radial integration.
This method provides two distinct advantages over traditional textbook derivations:

1. **Mathematical Simplicity:** It avoids complex three-dimensional coordinate transformations and the calculation of Jacobian determinants required in toroidal coordinates.

2. **Physical Insight:** It offers a clear, intuitive bridge between 2D surface geometry and 3D volume distribution by treating the solid object as a continuous collection of concentric hollow shells.

The result $I_{\text{solid}} = M\left(R^2 + \frac{3}{4}r^2\right)$ highlights how exploiting axial and geometric symmetries can significantly streamline rotational dynamics problems in classical mechanics.